

Electromagnetism 2

(spring semester 2026)

Lecture 10

Magnetic field in materials (part 1)

- ❖ Magnetised materials in non-uniform fields
- ❖ Surface and volume magnetisation currents
- ❖ The auxiliary H field
- ❖ Ampere–Maxwell law for the H field

Previous lecture

- ❖ External magnetic fields induce magnetisation of materials, which is characterised by the *magnetic susceptibility* χ_B :

$$\vec{M} = \chi_B \frac{\vec{B}}{\mu_0}$$

- ❖ *Diamagnetism* ($\chi_B < 0$) is a universal phenomenon: *induced magnetic dipole moments* are directed against the **B** field.
- ❖ *Paramagnetism* ($\chi_B > 0$) is due to alignment of *permanent atomic magnetic moments* along the **B** field.
- ❖ Microscopic theory of magnetism leads to

$$\chi_B = \mu_0 n \left(\frac{\mu^2}{3kT} - \frac{Ze^2 r_0^2}{6m_e} \right)$$

Magnetised materials in B-field

Material placed in a non-uniform magnetic field, $\vec{B}(z) = B(z) \cdot \vec{e}_z$

Magnetisation gives rise to a magnetic force acting on the sample in a non-uniform magnetic field:

$$F_{\text{mag}} = \mu n V \cdot \frac{dB}{dz} = M V \cdot \frac{dB}{dz} = \frac{\chi_B B}{\mu_0} V \cdot \frac{dB}{dz}$$

μ : induced magnetic moment of an atom [$\text{A} \cdot \text{m}^2$]

n : density of atoms [m^{-3}]

$M = \mu n$: magnetisation [A/m]

V : volume of the sample [m^3]

- ❖ Diamagnetics ($\chi_B < 0$) are repelled from regions of strong field.
- ❖ Paramagnetics ($\chi_B > 0$) are attracted into regions of strong field.

Levitation: magnetic force compensates gravity.

$$F_{\text{mag}} = mg = \rho V g ; \quad B \frac{dB}{dz} = \frac{\mu_0 \rho g}{|\chi_B|}$$

For water ($\chi_B \approx -10^{-5}$), we need $B \cdot dB/dz \approx 1300 \text{ T}^2/\text{m}$.
Much stronger effect for superconductors ($\chi_B = -1$).

A levitating frog
(Nijmegen, 2005)



Surface magnetisation currents

Consider a cylinder made of paramagnetic material, placed into a uniform magnetic field, parallel to the field lines.

Magnetic dipole moment of a horizontal slice δx :

$$m = \underbrace{\pi r^2 M \delta x}_{\text{By definition of } \mathbf{M}, \text{ magnetisation x volume}} = \underbrace{\pi r^2 i_M \delta x}_{\text{By definition of } \mathbf{m}, \text{ area x current (lecture 5)}}$$

By definition of $\mathbf{M}$,
magnetisation x volume

By definition of $\mathbf{m}$,
area x current
(lecture 5)

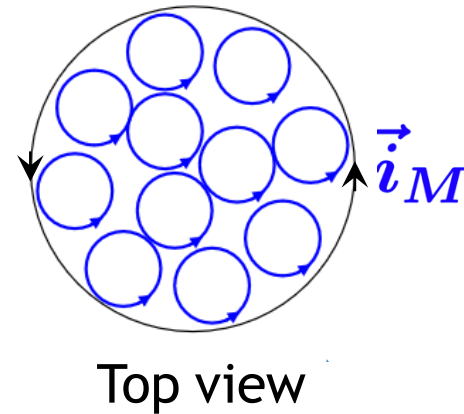
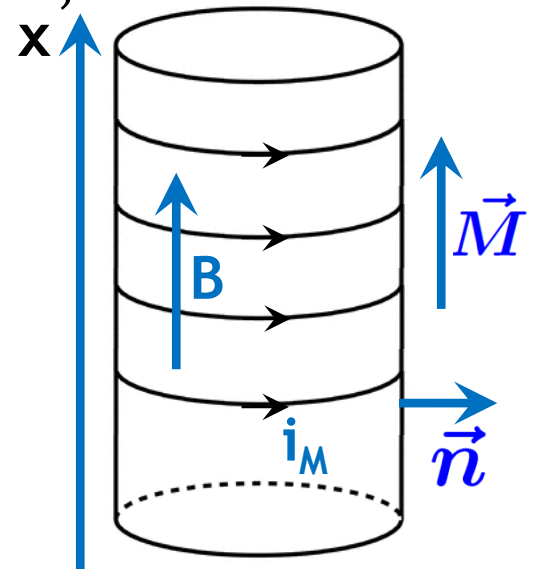
Surface current density: $i_M = M$ [A/m]

In the general case, $\boxed{\vec{i}_M = \vec{M} \times \vec{n}}$

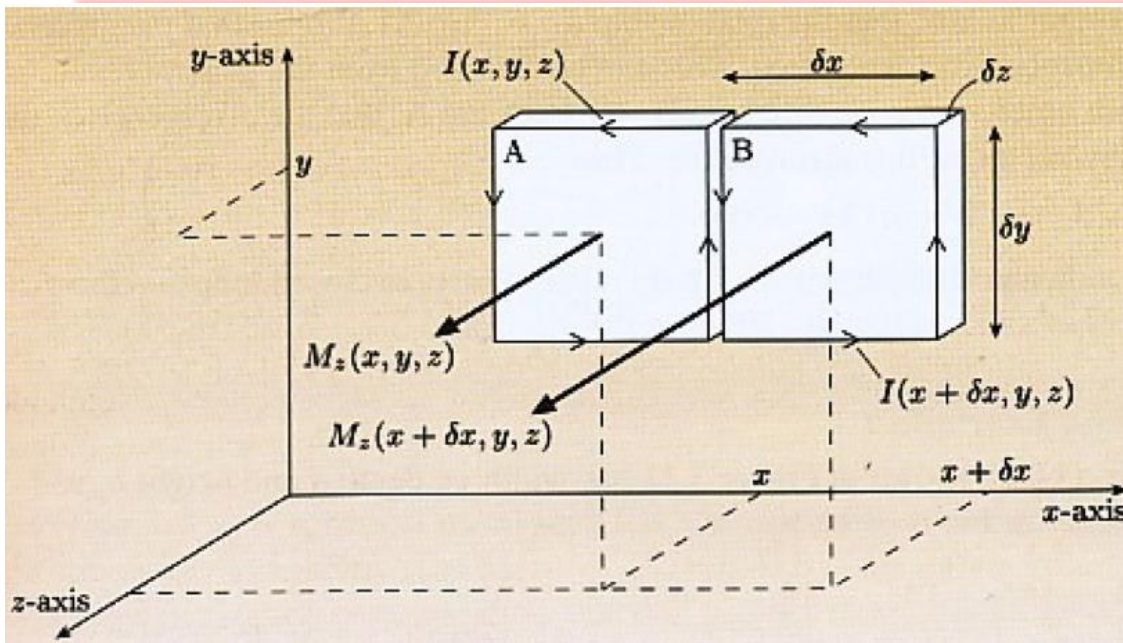
$\vec{n}$: unit normal vector pointing outwards

❖ Surface currents are perpendicular to $\mathbf{M}$.

❖ Currents flow on any surface which is **not** perpendicular to $\mathbf{M}$. 3



Volume magnetisation currents (1)



Volume currents flow in *non-uniformly magnetised* materials.

Assume magnetisation $\vec{M}$ along the **z** axis.

Consider two adjacent identical small volumes A, B.

Magnetic moments of the volumes:

$$m_A = M_z(x, y, z) \cdot dxdydz$$

$$m_B = M_z(x + dx, y, z) \cdot dxdydz = m_A + \left(\frac{\partial M_z}{\partial x} dx \right) \cdot dxdydz$$

Current in loop B:

$$I_B = \frac{m_B}{dxdy} = \frac{m_A + \frac{\partial M_z}{\partial x} dx \cdot dxdydz}{dxdy} = I_A + \frac{\partial M_z}{\partial x} dx dz$$

Volume magnetisation currents (2)

Net current in the y direction at the boundary between two volumes:

$$I_y = I_A - I_B = -\frac{\partial M_z}{\partial x} dx dz$$

Electric current density, by definition: $j_y = \frac{I_y}{dx dz} = -\frac{\partial M_z}{\partial x}$

There is similar contribution to j_y from the dependence of the magnetisation component M_x on z , therefore

$$j_y = \frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x}$$

Similarly for the other components of $\vec{j}_M$:

$$j_x = \frac{\partial M_z}{\partial y} - \frac{\partial M_y}{\partial z} \quad \text{and} \quad j_z = \frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y}$$

Finally, $\vec{j}_M = \nabla \times \vec{M}$

Analogy with the electric field

Magnetisation of the sample, $\mathbf{M}(\mathbf{x},\mathbf{y},\mathbf{z})$, can be replaced by an equivalent distribution of *magnetisation currents*:

$$\vec{i}_M = \vec{M} \times \vec{n} \qquad \vec{j}_M = \nabla \times \vec{M}$$

This is similar to the electric field (*lecture 7*):

polarisation of the sample, $\mathbf{P}(\mathbf{x},\mathbf{y},\mathbf{z})$, can be replaced by an equivalent distribution of *polarisation charges*:

$$\sigma_p = \vec{P} \cdot \vec{n} \qquad \rho_p = -\nabla \cdot \vec{P}$$

Let's introduce an auxiliary magnetic field, $\mathbf{H}$, to facilitate field calculations, similarly to the electric displacement field $\mathbf{D}$.

The H-field; Ampere–Maxwell law

Maxwell's equation **M4** (*lecture 6*); the current is the sum of conduction (“free”), polarisation and magnetisation currents:

$$\nabla \times \vec{B} = \mu_0(\vec{j}_f + \vec{j}_p + \vec{j}_M) + \epsilon_0\mu_0 \frac{\partial \vec{E}}{\partial t} =$$

$$\mu_0\vec{j}_f + \underbrace{\mu_0 \frac{\partial \vec{P}}{\partial t}}_{\text{See lecture 7}} + \mu_0 \nabla \times \vec{M} + \epsilon_0\mu_0 \frac{\partial \vec{E}}{\partial t} = \mu_0\vec{j}_f + \mu_0 \frac{\partial \vec{D}}{\partial t} + \mu_0 \nabla \times \vec{M}$$

Therefore $\nabla \times \left(\frac{\vec{B}}{\mu_0} - \vec{M} \right) = \vec{j}_f + \frac{\partial \vec{D}}{\partial t}$ $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ (*lecture 8*)

Let's define an auxiliary **H** field: $\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$

Then $\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$

which replaces **(M4)**, accounting for electric and magnetic properties of the medium (the index “f” is implied)

Displacement current density in matter (*see also lecture 6*)

Summary

- ❖ Volume distribution of magnetisation $\mathbf{M}$ can be replaced by an equivalent distribution of *magnetisation currents*:

$$\vec{i}_M = \vec{M} \times \vec{n} \quad \vec{j}_M = \nabla \times \vec{M}$$

- ❖ The auxiliary $\mathbf{H}$ -field is defined as

$$\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}, \text{ or equivalently } \vec{B} = \mu_0(\vec{H} + \vec{M})$$

- ❖ In dielectric and magnetic media, Maxwell's equation (M4) becomes

$$\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$$